

micrOS platform analytics

A guide to platform evolution, dependencies and engineering health

1.48.0-1 to 3.7.0-0 / 189 recorded versions

Explore how the runtime grows, which files it depends on, and how the project performs.

Evolution

Core, load module and package growth across versions.
Compare file counts and source lines over time.

Dependencies

Core reference histories, ranked by latest count.
Lower-reference files are listed separately.

People

Contribution shares and contributor areas.
See who contributes and which files they work on.

Quality

Pylint scores and load module dependency warnings.
Track code quality and dependency checks over time.

Devices

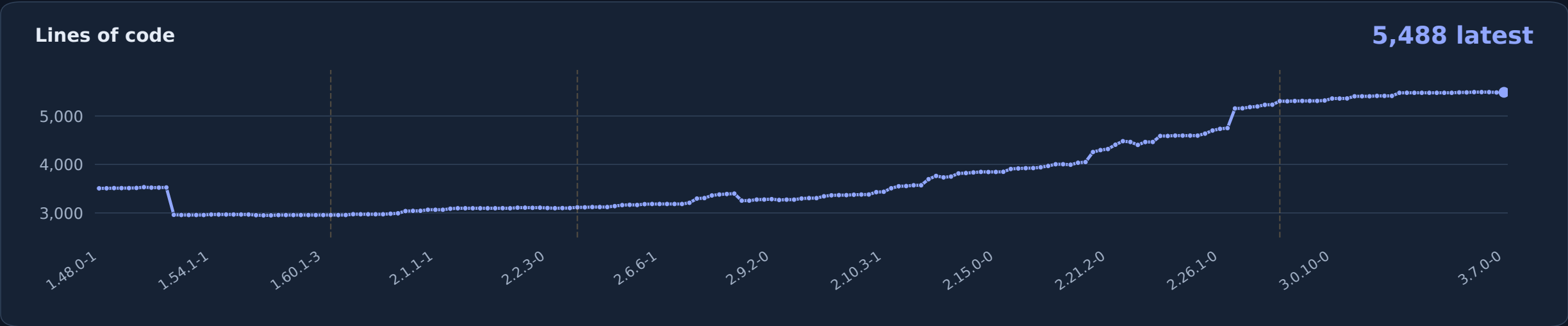
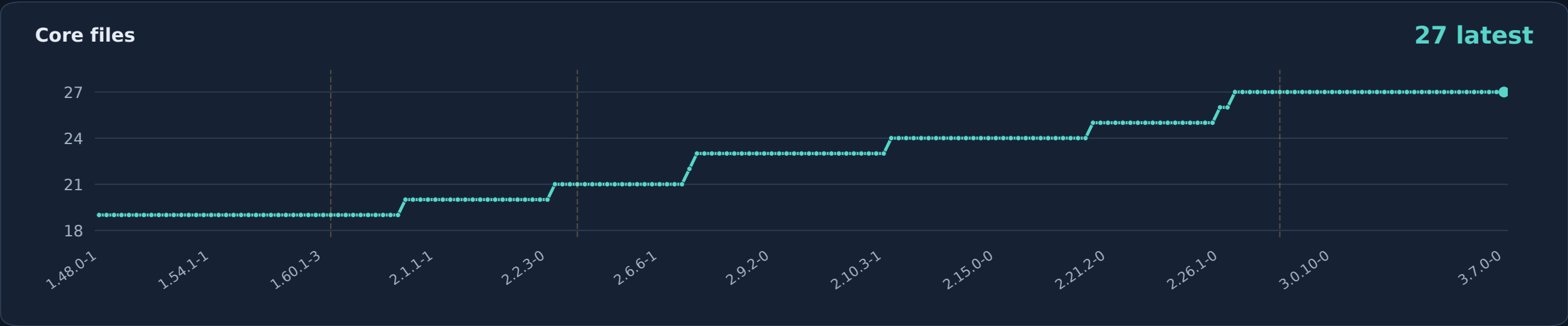
Recorded device benchmarks and module inventories.
Compare measured performance and loaded modules.

History

Recent version commits, with newest entries first.
Read the full commit messages at the end of the report.

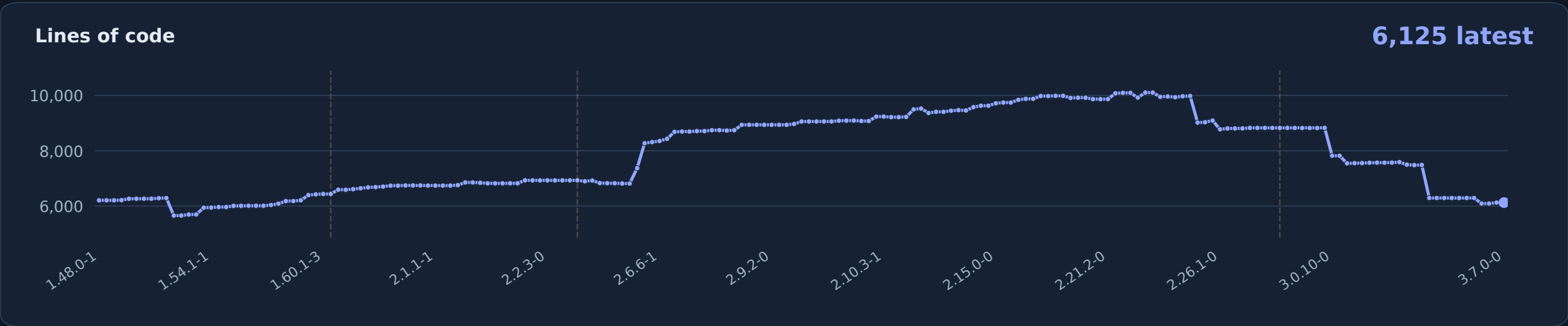
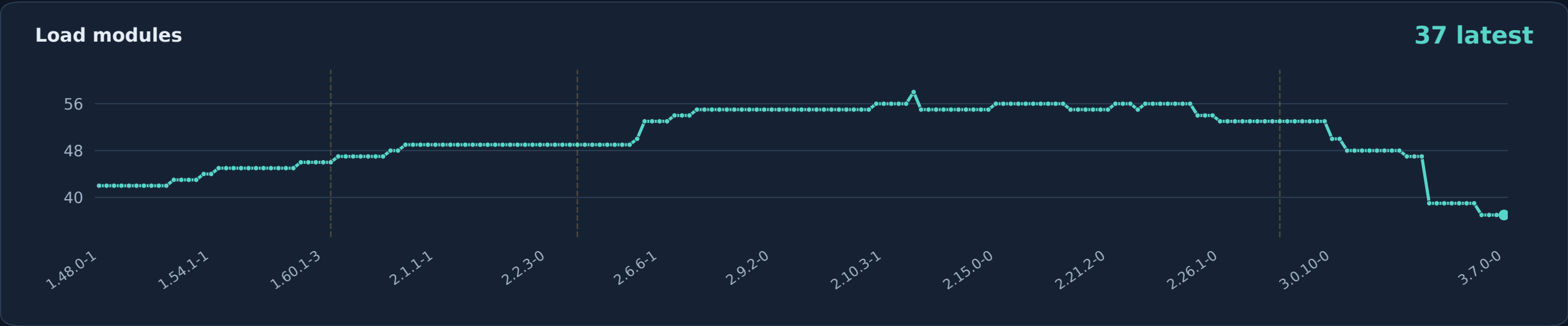
Core system evolution

1.48.0-1 to 3.7.0-0 / 189 recorded versions



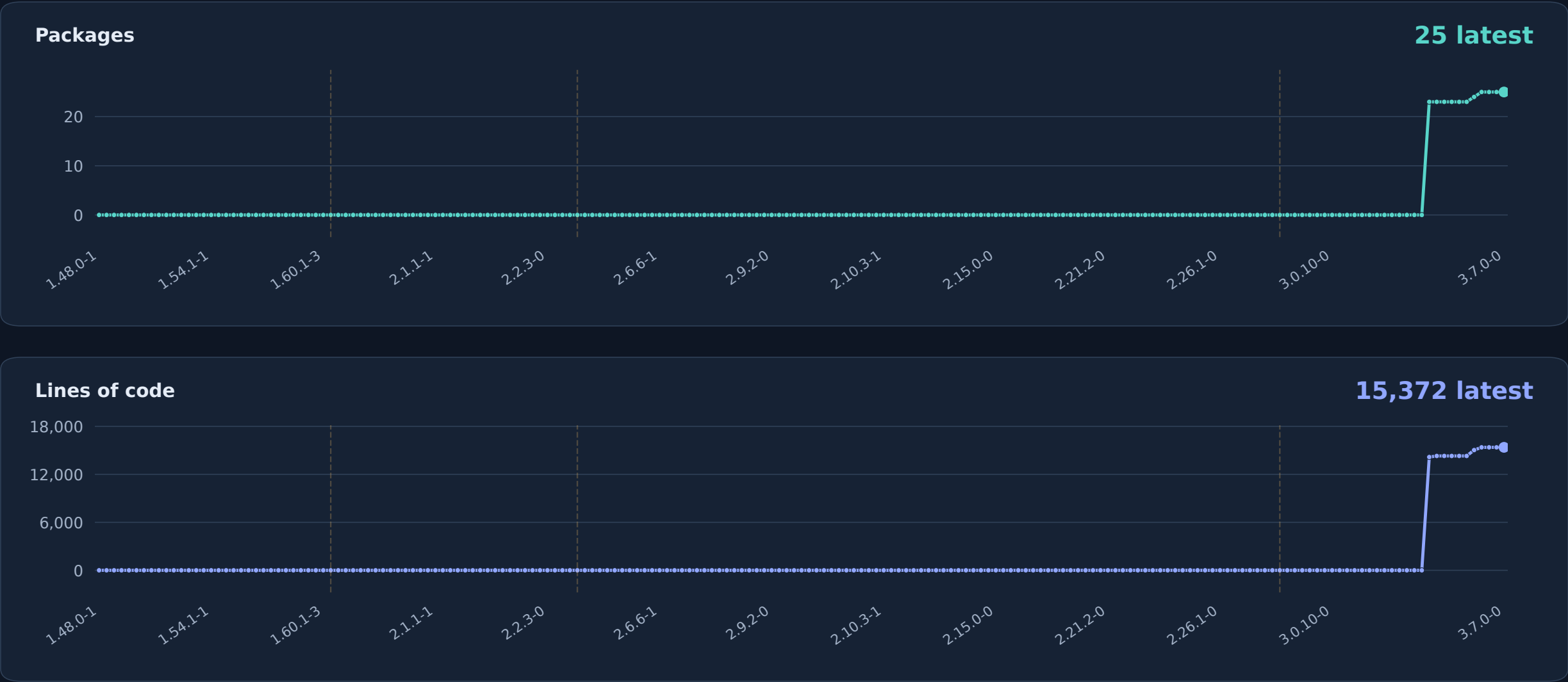
Load module evolution

1.48.0-1 to 3.7.0-0 / 189 recorded versions



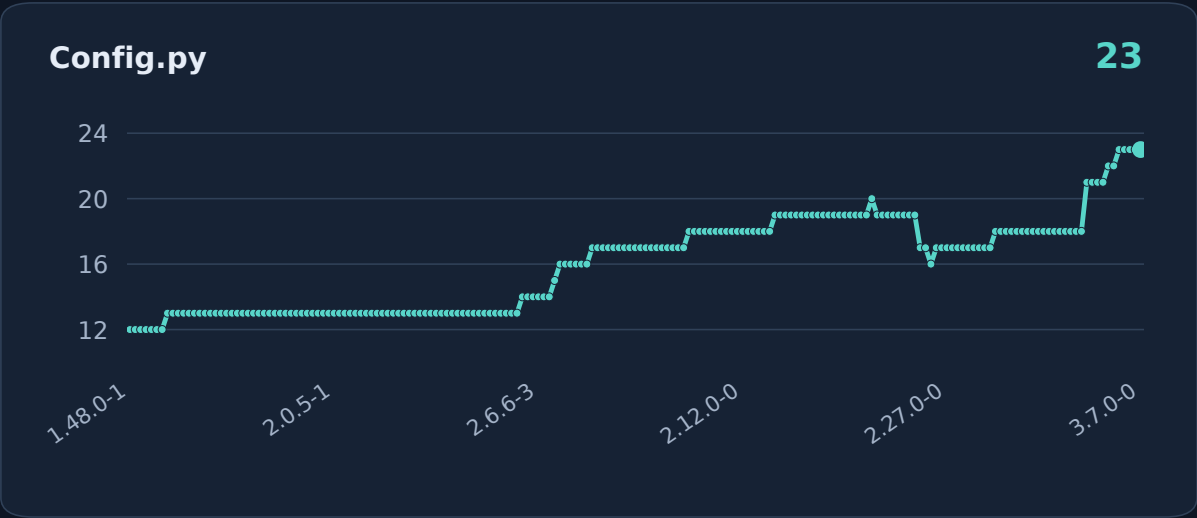
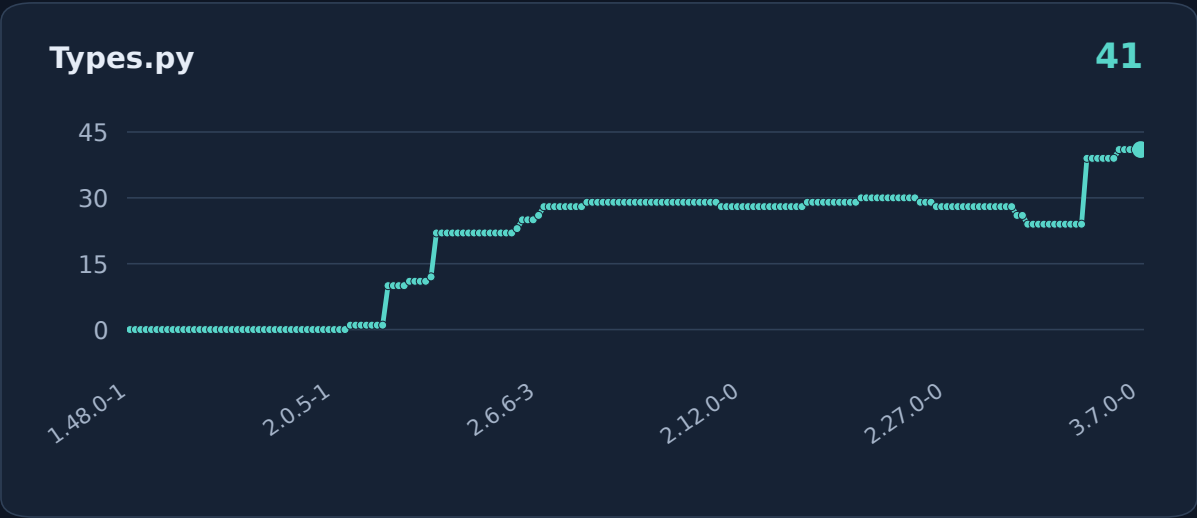
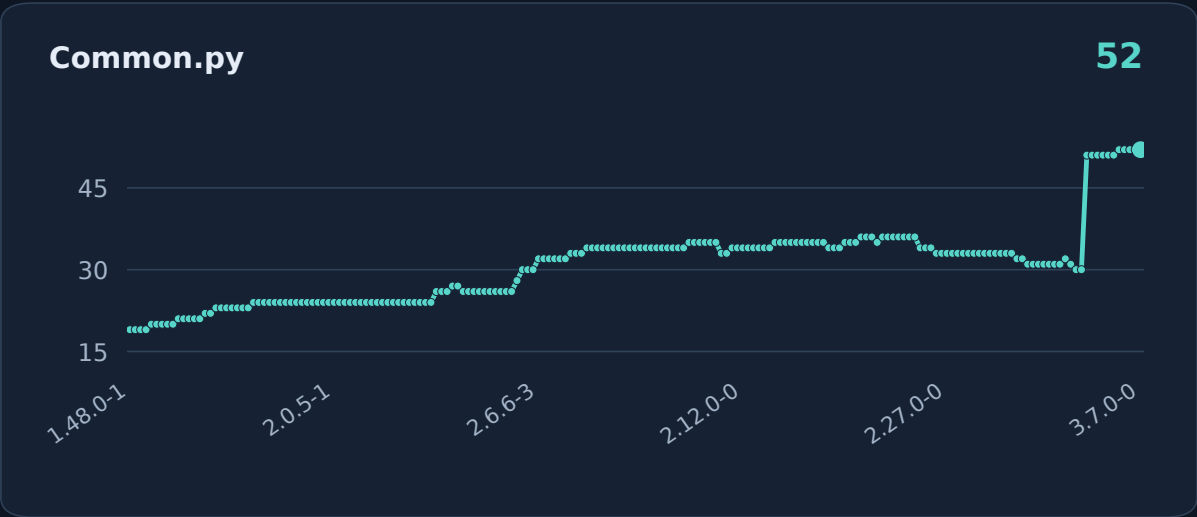
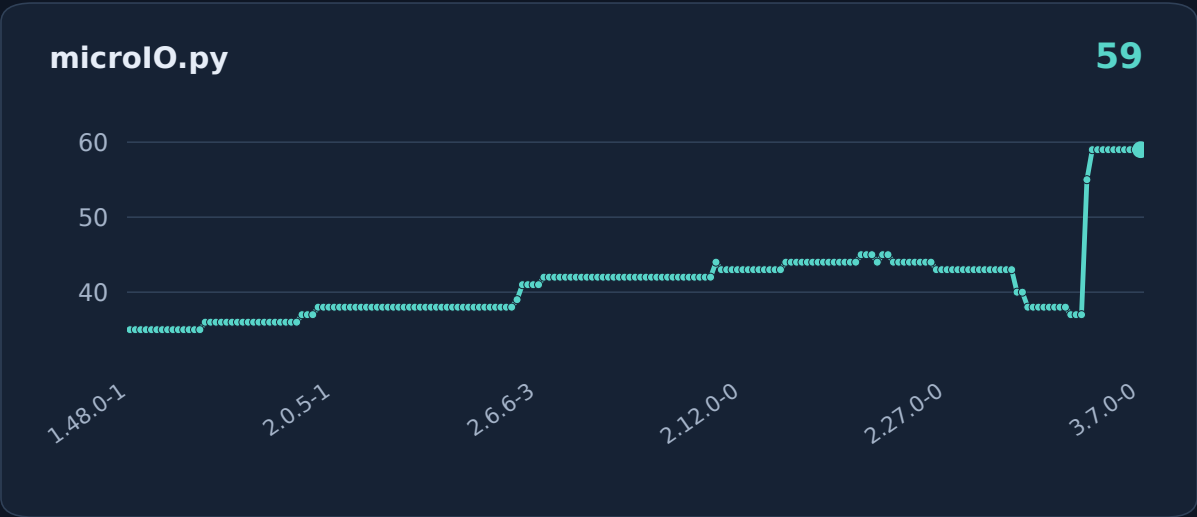
Package evolution

1.48.0-1 to 3.7.0-0 / 189 recorded versions



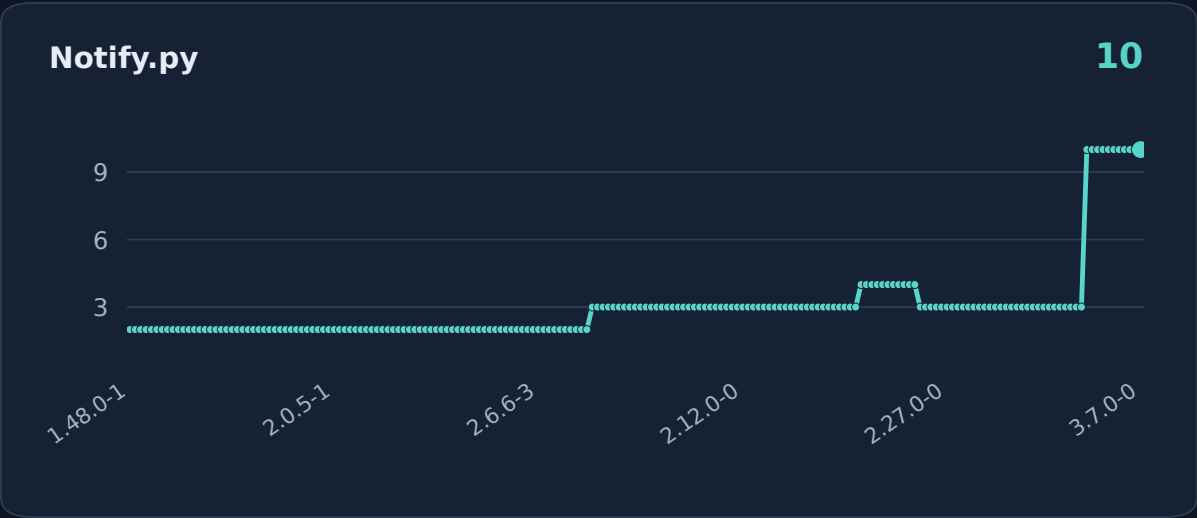
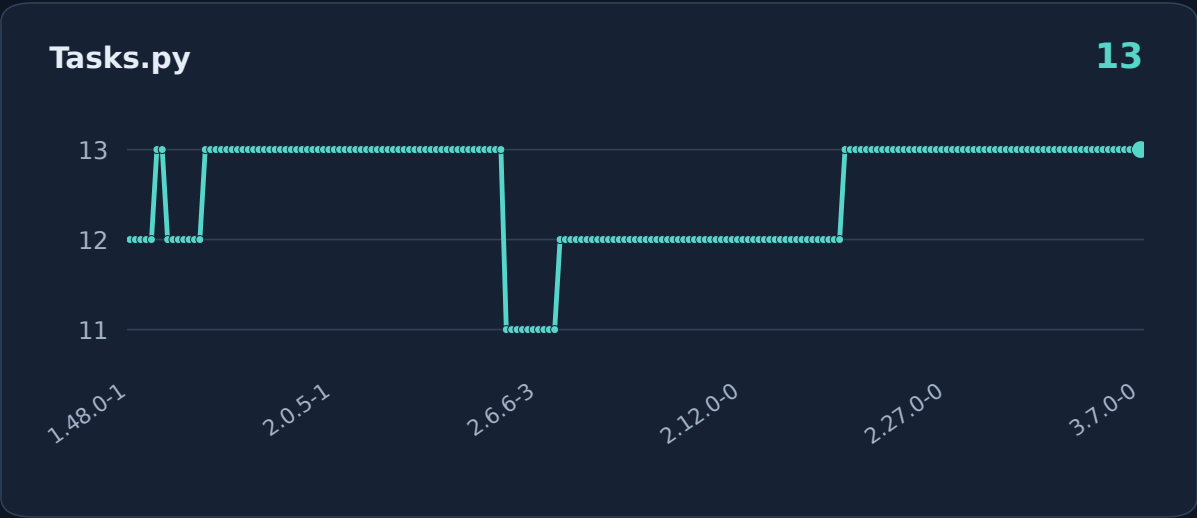
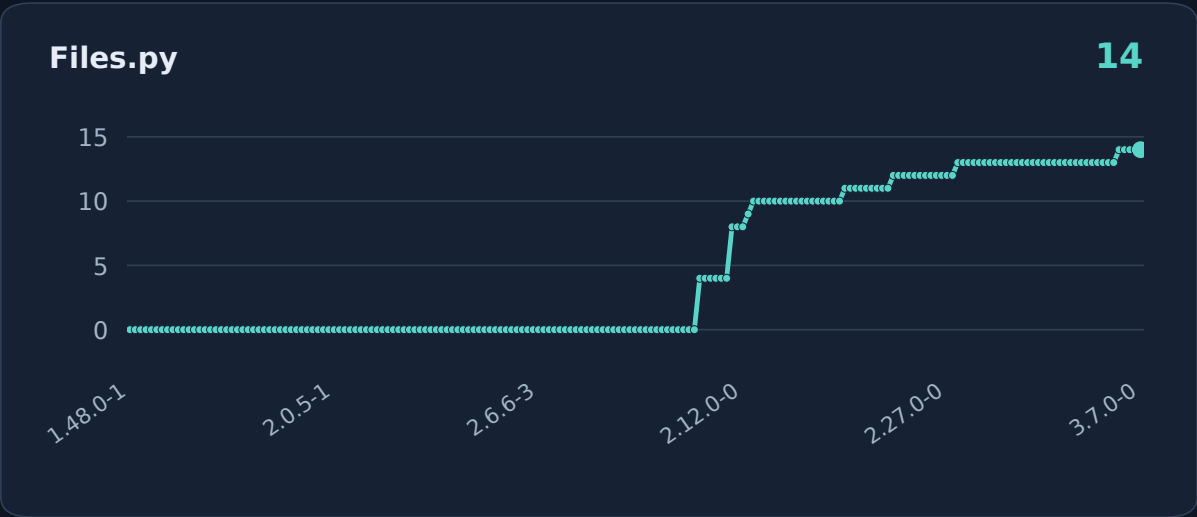
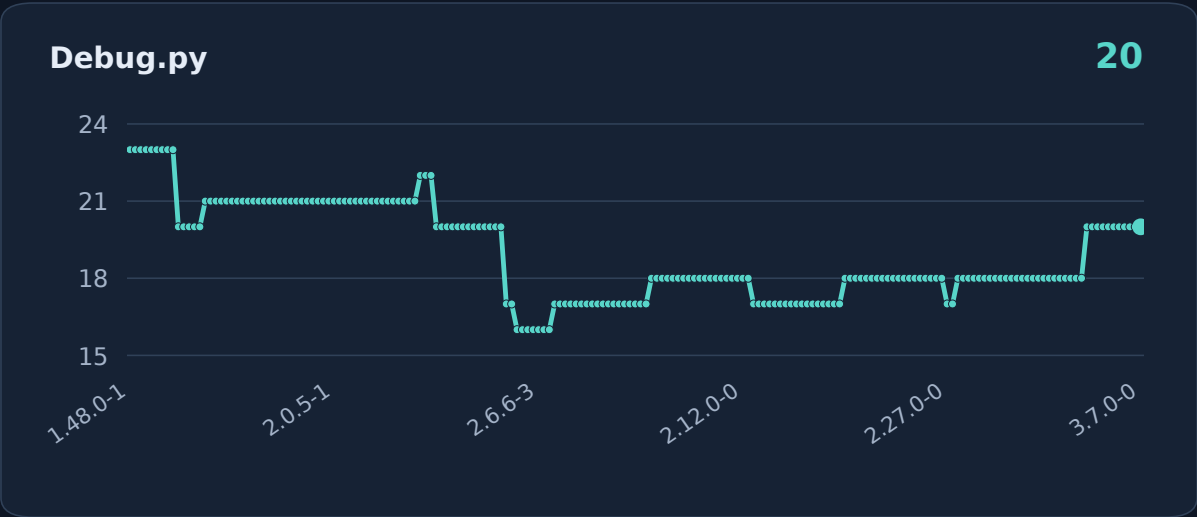
Core references

Per-file histories / latest reference count above 3



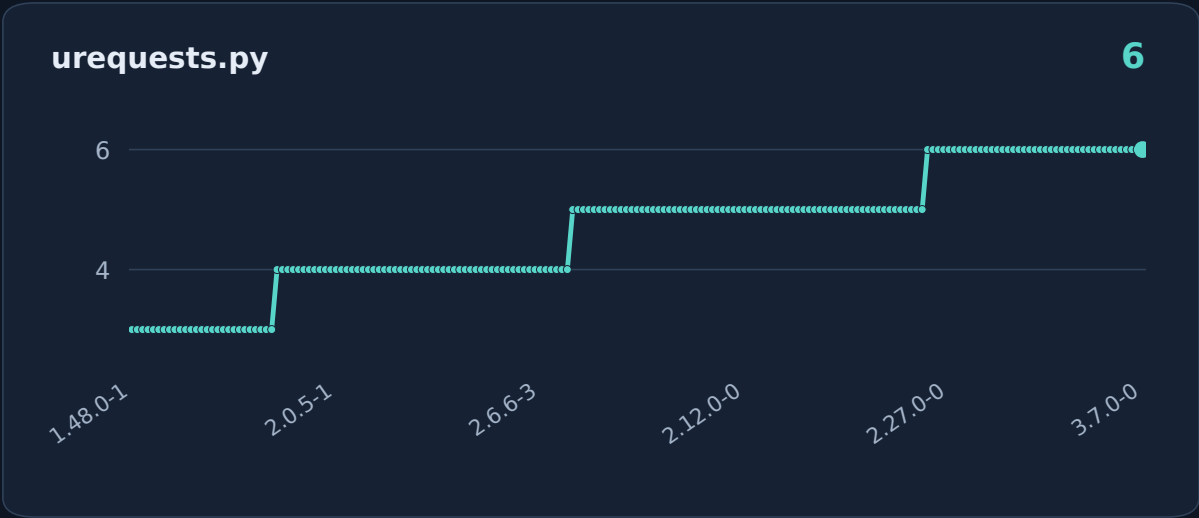
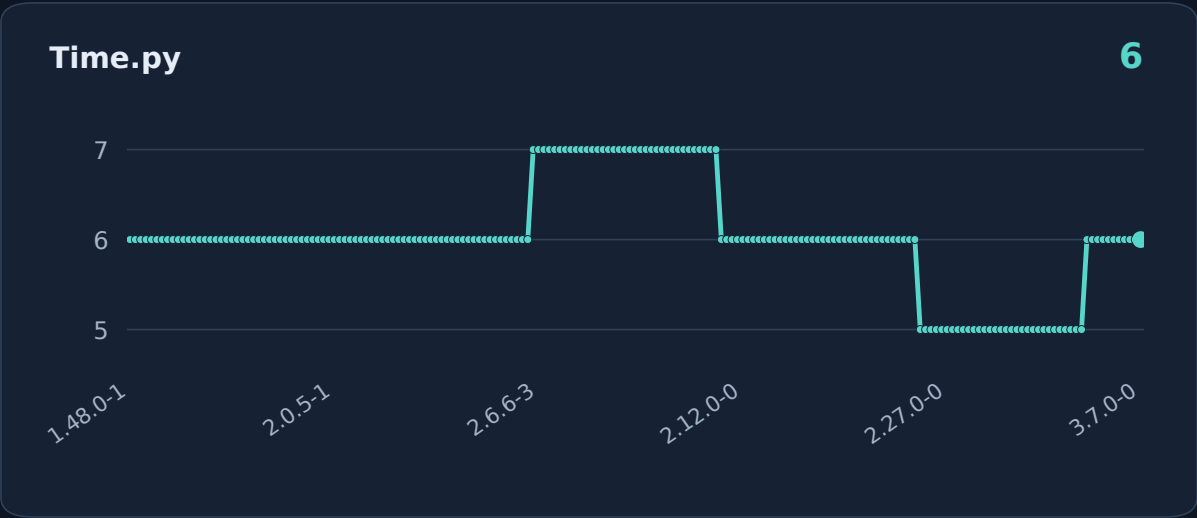
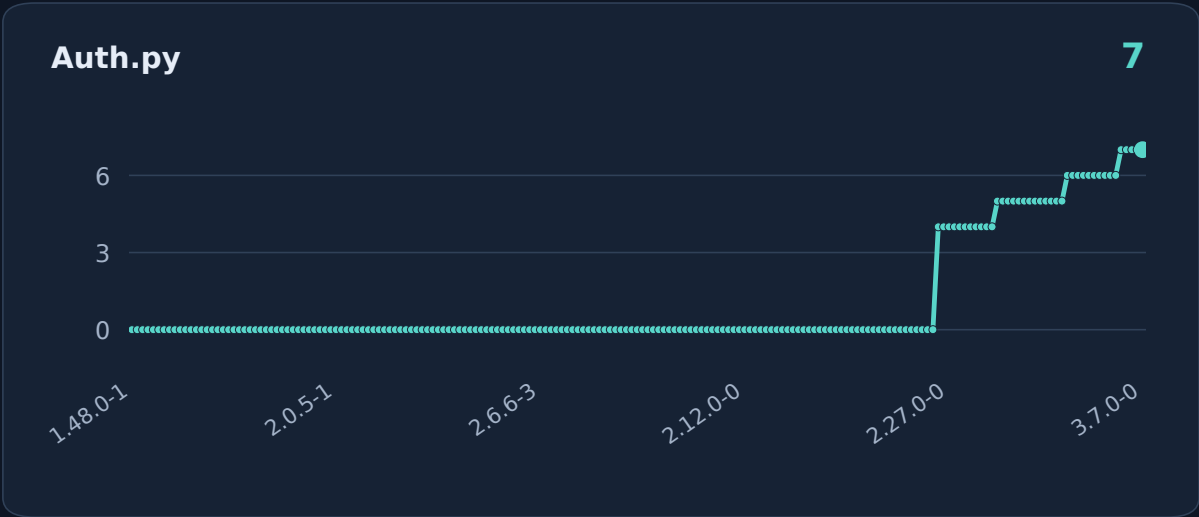
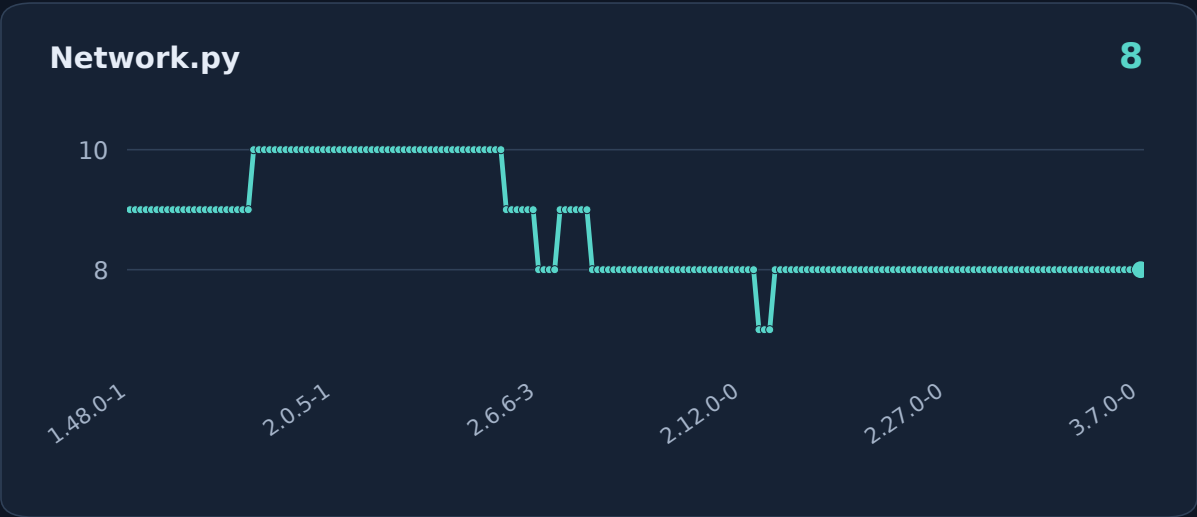
Core references

Per-file histories / latest reference count above 3



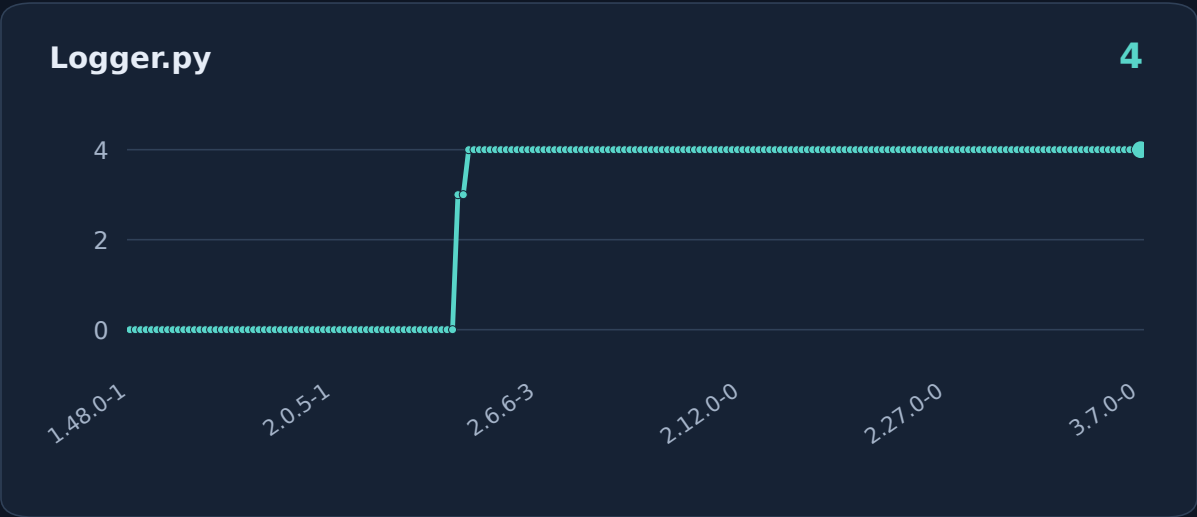
Core references

Per-file histories / latest reference count above 3



Core references

Per-file histories / latest reference count above 3



Lower-reference core files

Latest count of 3 or fewer / excluded from the trend charts

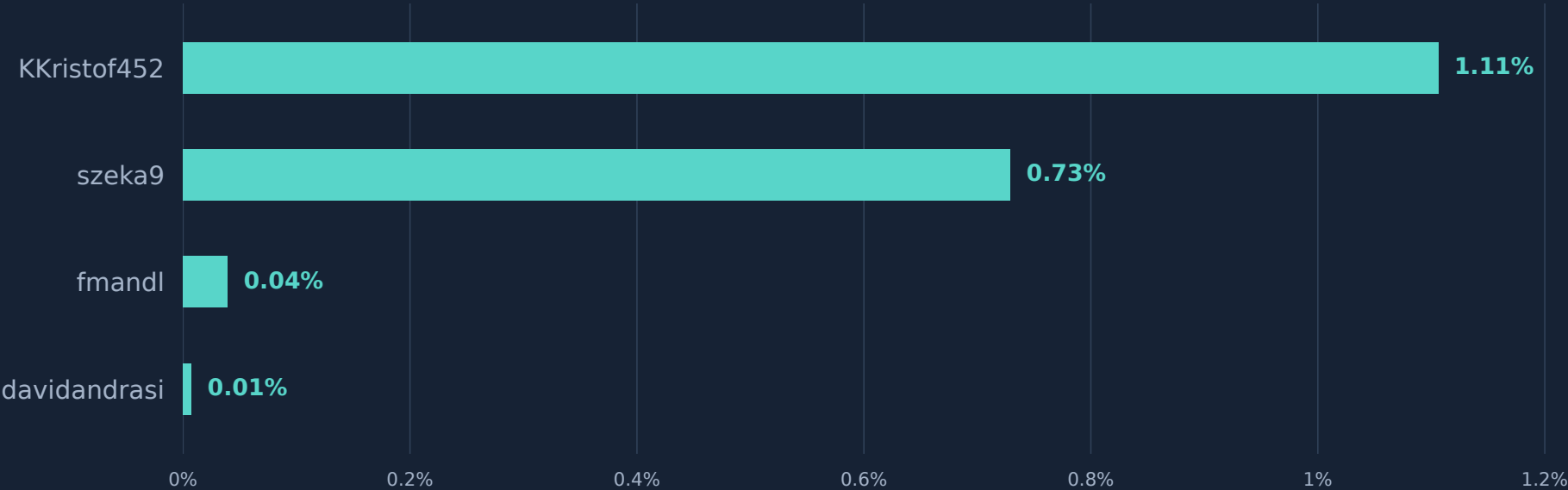
File	Latest references
Pacman.py	3
Server.py	3
InterConnect.py	2
Shell.py	2
Web.py	2
Buffer.py	1
Hooks.py	1
Interrupts.py	1
Scheduler.py	1
micrOS.py	1
micrOSloader.py	1
Espnow.py	0
helpers.py	0
main.py	0
reset.py	0

Project contributors

Share of project contributions / lead contributors shown above

BNM

98.12%



The bar chart shows contributors with a share of 30% or less.

Contributors' file changes

Complete recorded file lists / long paths wrap and continue onto the next card

KKristof452 / 56 file entries

apps/SysCheck_app.py
micrOS/ConfigHandler.py
micrOS/LM_L298N_DCmotor.py
micrOS/LM_system.py
micrOS/LM_system.py
micrOS/Network.py
micrOS/Scheduler.py
micrOS/SocketServer.py
micrOS/Time.py
micrOS/Time.py
micrOS/source/Espnow.py
micrOS/source/Espnow.py
micrOS/source/InterConnect.py
micrOS/source/InterConnect.py
micrOS/source/InterpreterShell.py
micrOS/source/LM_ds18.py
micrOS/source/LM_keychain.py
micrOS/source/LM_neoeffects.py
micrOS/source/LM_neopixel.py
micrOS/source/LM_oled_ui.py
micrOS/source/LM_rgb.py
micrOS/source/LM_system.py
micrOS/source/Scheduler.py
micrOS/source/Shell.py
micrOS/source/Shell.py
micrOS/source/SocketServer.py
micrOS/source/Tasks.py

KKristof452 / 56 file entries / continued

micrOS/source/Tasks.py
micrOS/source/Time.py
micrOS/source/Web.py
micrOS/source/dashboard.html
micrOS/source/micrOS.py
micrOS/source/modules/LM_espnw.py
micrOS/source/modules/LM_oled_ui.py
micrOS/source/modules/LM_oledui.py
micrOS/source/modules/LM_web.py
micrOS/source/udashboard.js
micrOS/source/urequests.py
micrOS/source/ustyle.css
micrOS/source/uwidgets.js
micrOS/source/web/config.css
micrOS/source/web/config.html
micrOS/source/web/config.js
micrOS/source/web/index.html
micrOS/source/web/uapi.js
micrOS/source/web/uapi_widgets.js
micrOS/source/web/ustyle.css
micrOS/utests/test_web_engine.py
toolkit/DevEnvCompile.py
toolkit/dashboard_apps/uLightDemo.py
toolkit/workspace/precompiled/modules/LM_espnw.py
toolkit/workspace/precompiled/web/config.css
toolkit/workspace/precompiled/web/config.html
toolkit/workspace/precompiled/web/config.js

Contributors' file changes

Complete recorded file lists / long paths wrap and continue onto the next card

KKristof452 / 56 file entries / continued

toolkit/workspace/precompiled/web/uapi.js
toolkit/workspace/precompiled/web/ustyle.css

szeka9 / 36 file entries

APPLICATION_GUIDE.md
README.md
env/requirements.txt
media/lms/i2s_mic.png
micrOS/source/Buffer.py
micrOS/source/Common.py
micrOS/source/Config.py
micrOS/source/LM_OV2640.py
micrOS/source/LM_aht10.py
micrOS/source/LM_i2s_mic.py
micrOS/source/LM_presence.py
micrOS/source/LM_sound_event.py
micrOS/source/Server.py
micrOS/source/Shell.py
micrOS/source/Tasks.py
micrOS/source/Web.py
micrOS/source/modules/LM_OV2640.py
micrOS/source/modules/LM_fileserver.py
micrOS/source/modules/LM_i2s_mic.py
micrOS/source/web/editor.js
micrOS/source/web/filesui.html
micrOS/source/web/filesui.js
micrOS/utests/test_buffer.py
micrOS/utests/test_web_engine.py
setup.py
toolkit/LM_to_compile.dat
toolkit/dashboard_apps/MicrophoneTest_app.py

Contributors' file changes

Complete recorded file lists / long paths wrap and continue onto the next card

szeka9 / 36 file entries / continued

```
toolkit/dashboard_apps/PresenceTest.py
toolkit/dashboard_apps/_capture.py
toolkit/dashboard_apps/_micPlotting.py
toolkit/dashboard_apps/_presencePlotting.py
toolkit/simulator_lib/machine.py
toolkit/workspace/precompiled/LM_presence.py
toolkit/workspace/precompiled/web/editor.js
toolkit/workspace/precompiled/web/filesui.html
toolkit/workspace/precompiled/web/filesui.js
```

fmandl / 27 file entries

```
micrOS/packages/alarm_system/package/LM_alarm_system.py
micrOS/packages/alarm_system/package/__init__.py
micrOS/packages/alarm_system/package/alarm_system/__init__.py
micrOS/packages/alarm_system/package/alarm_system/actions.py
micrOS/packages/alarm_system/package/alarm_system/door_sensor.py
micrOS/packages/alarm_system/package/alarm_system/event_log.py
micrOS/packages/alarm_system/package/alarm_system/mqtt_watcher.py
micrOS/packages/alarm_system/package/alarm_system/sms_handler.py
micrOS/packages/alarm_system/package/pacman.json
micrOS/packages/garage_remote/package/LM_garage.py
micrOS/packages/garage_remote/package/__init__.py
micrOS/packages/garage_remote/package/pacman.json
micrOS/packages/phone_manager/package/LM_users.py
micrOS/packages/phone_manager/package/__init__.py
micrOS/packages/phone_manager/package/manager.py
micrOS/packages/phone_manager/package/pacman.json
micrOS/packages/sim800/package/LM_sim800.py
micrOS/packages/sim800/package/LM_sim800_http.py
micrOS/packages/sim800/package/LM_sim800mqtt.py
micrOS/packages/sim800/package/__init__.py
micrOS/packages/sim800/package/codec.py
micrOS/packages/sim800/package/modem.py
micrOS/packages/sim800/package/notify.py
micrOS/packages/sim800/package/pacman.json
micrOS/source/LM_L298N.py
micrOS/source/LM_mqtt_client.py
micrOS/source/modules/LM_mqtt_client.py
```

Contributors' file changes

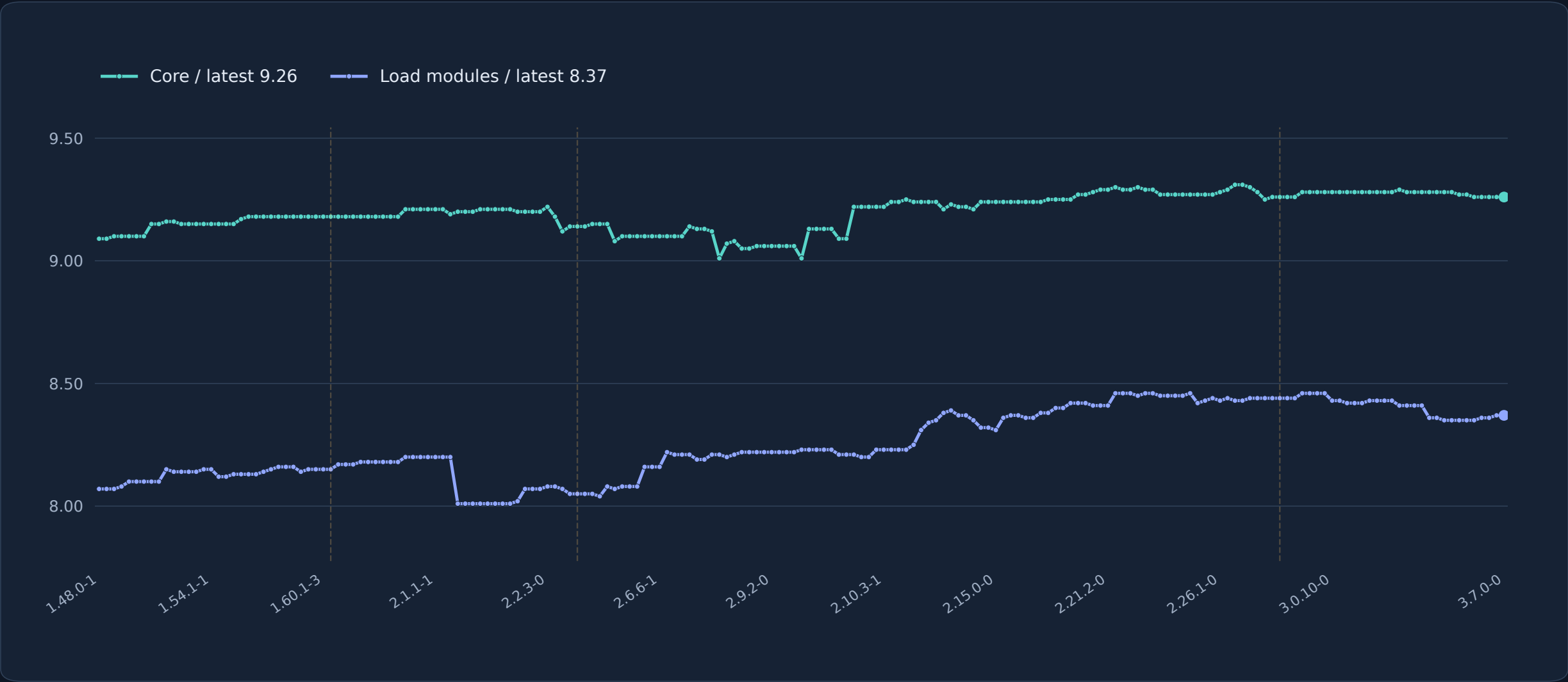
Complete recorded file lists / long paths wrap and continue onto the next card

davidandrasi / 3 file entries

tools/MicrOSDevEnv/MicrOSDevEnv.py
tools/MicrOSDevEnv/SearchDevices.py
tools/requirements.txt

Code quality

Pylint scores across every recorded version



Dependency warnings

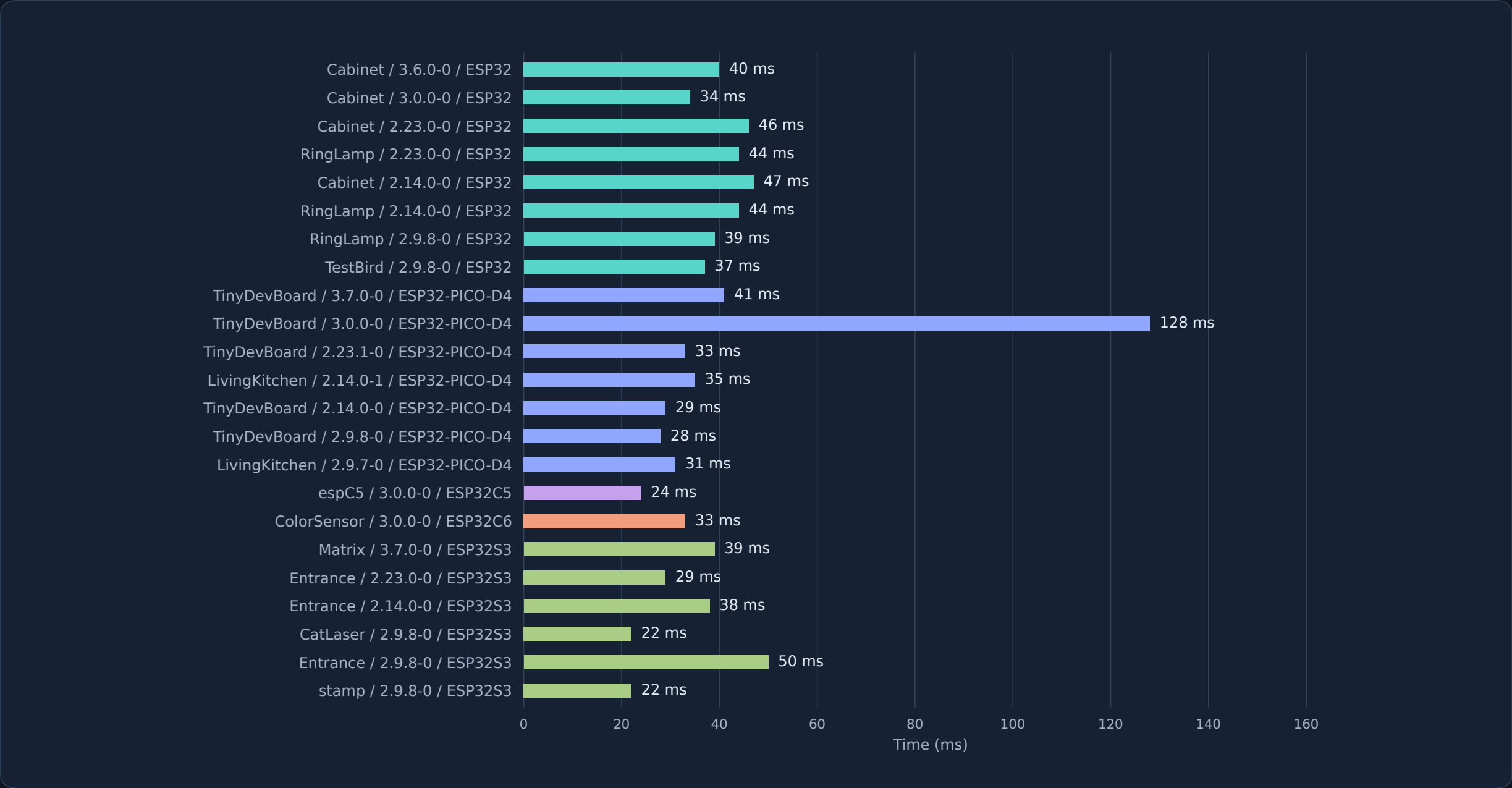
Load module dependency checks across every recorded version

5 latest warnings



Shell heartbeatx10 response

Device benchmarks / 23 records / newest versions first within each device type



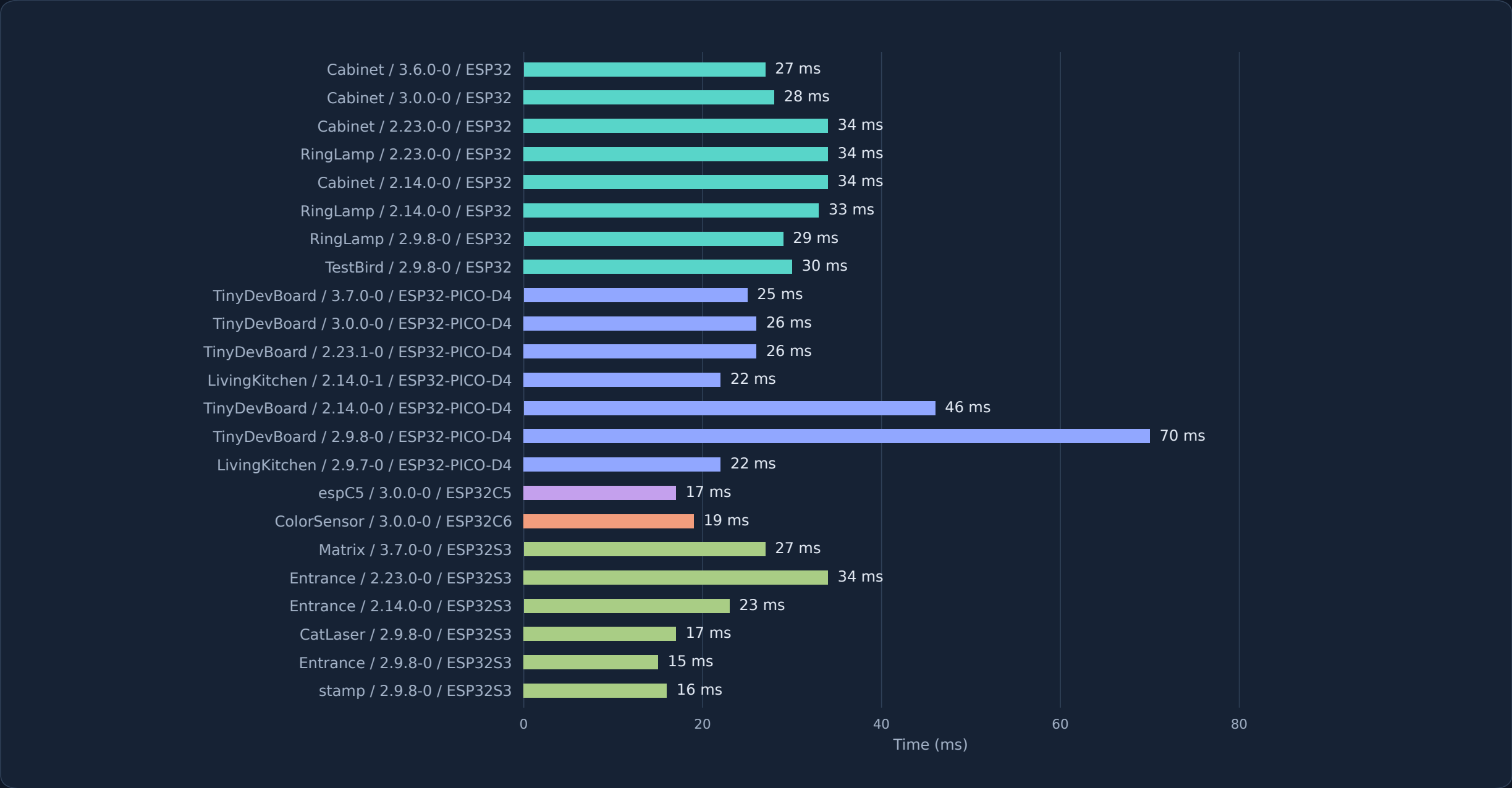
Shell multi session dt

Device benchmarks / 23 records / newest versions first within each device type



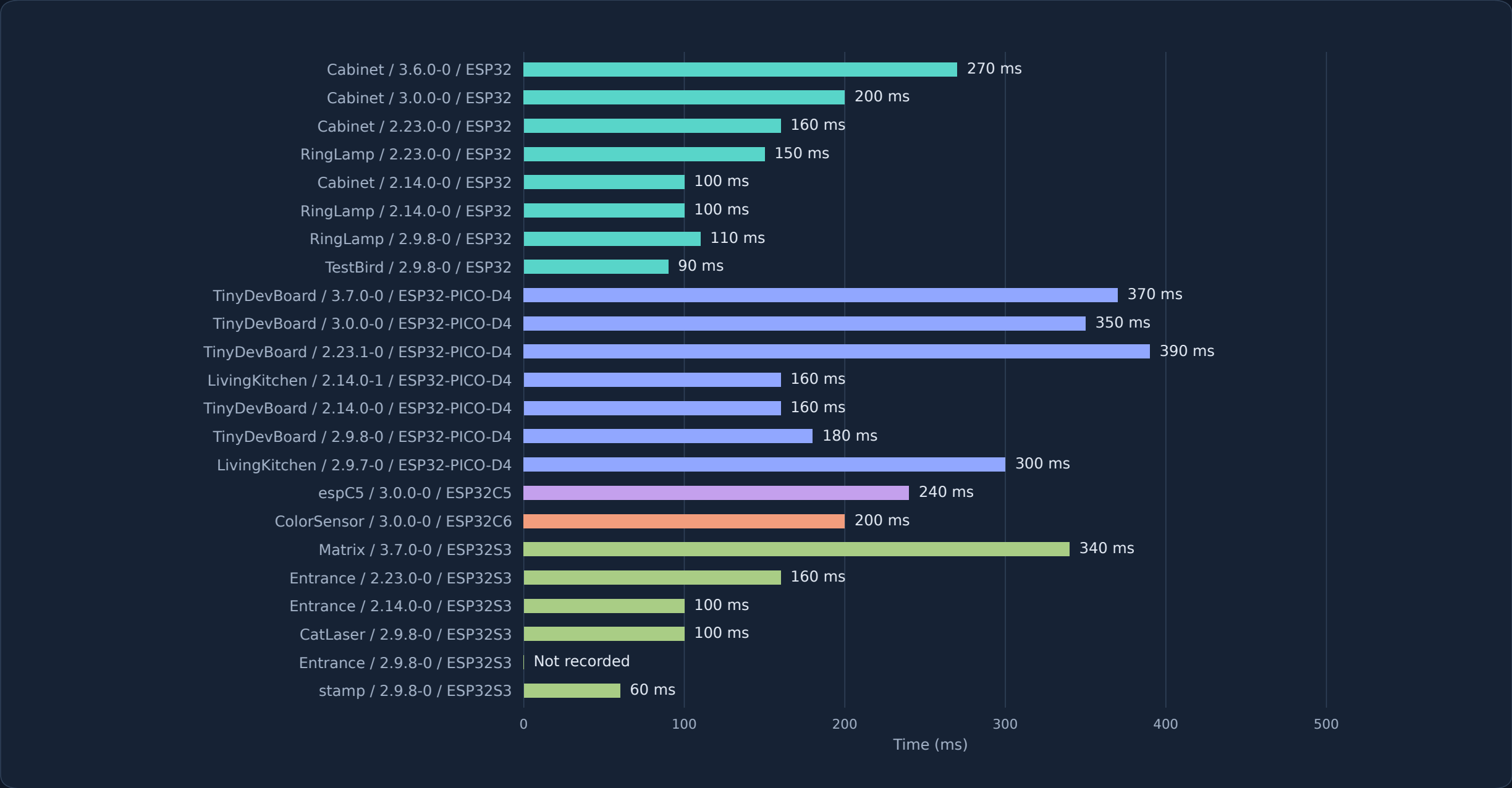
Shell single session dt

Device benchmarks / 23 records / newest versions first within each device type



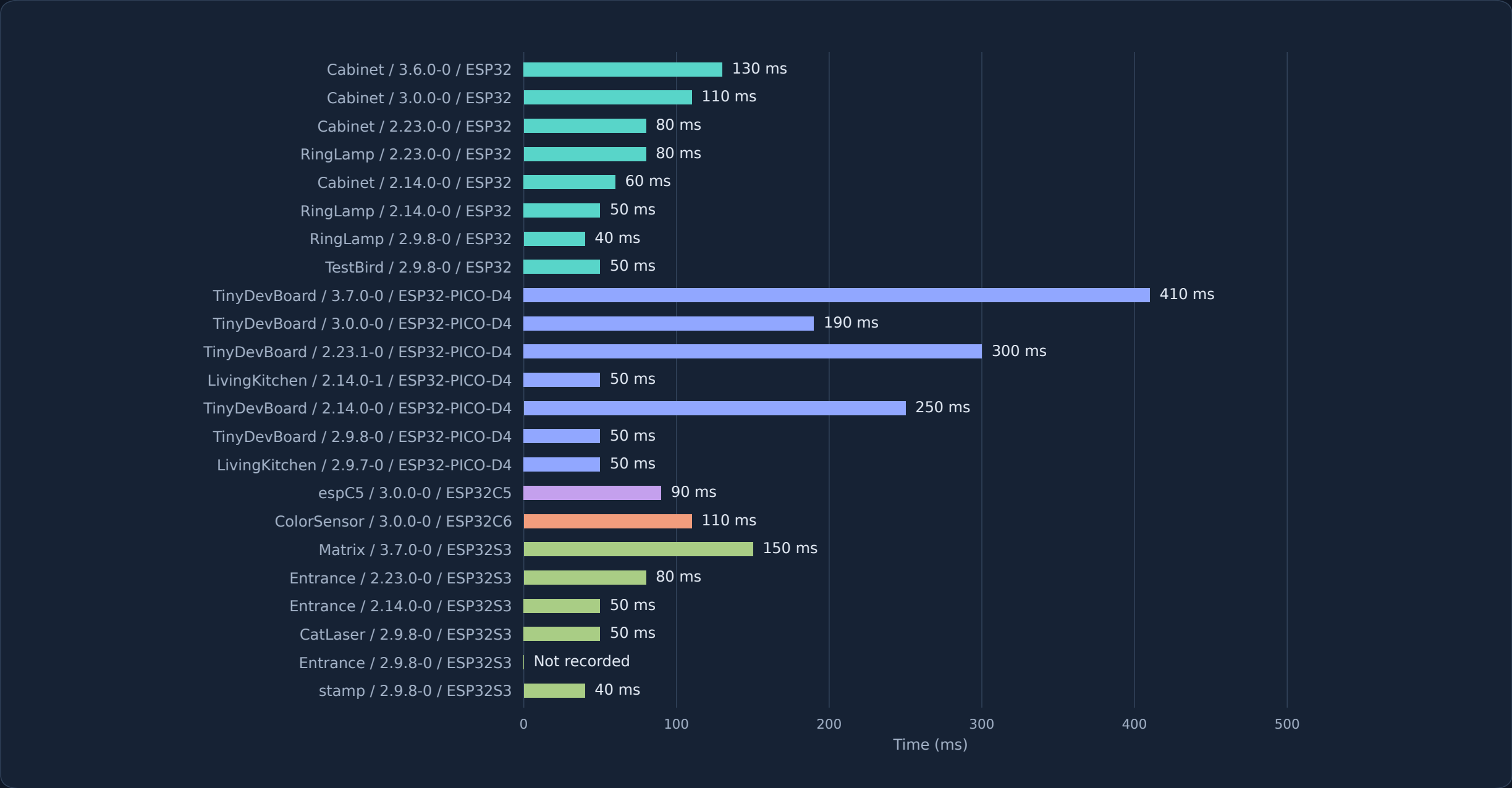
Web landingpage response

Device benchmarks / 23 records / newest versions first within each device type



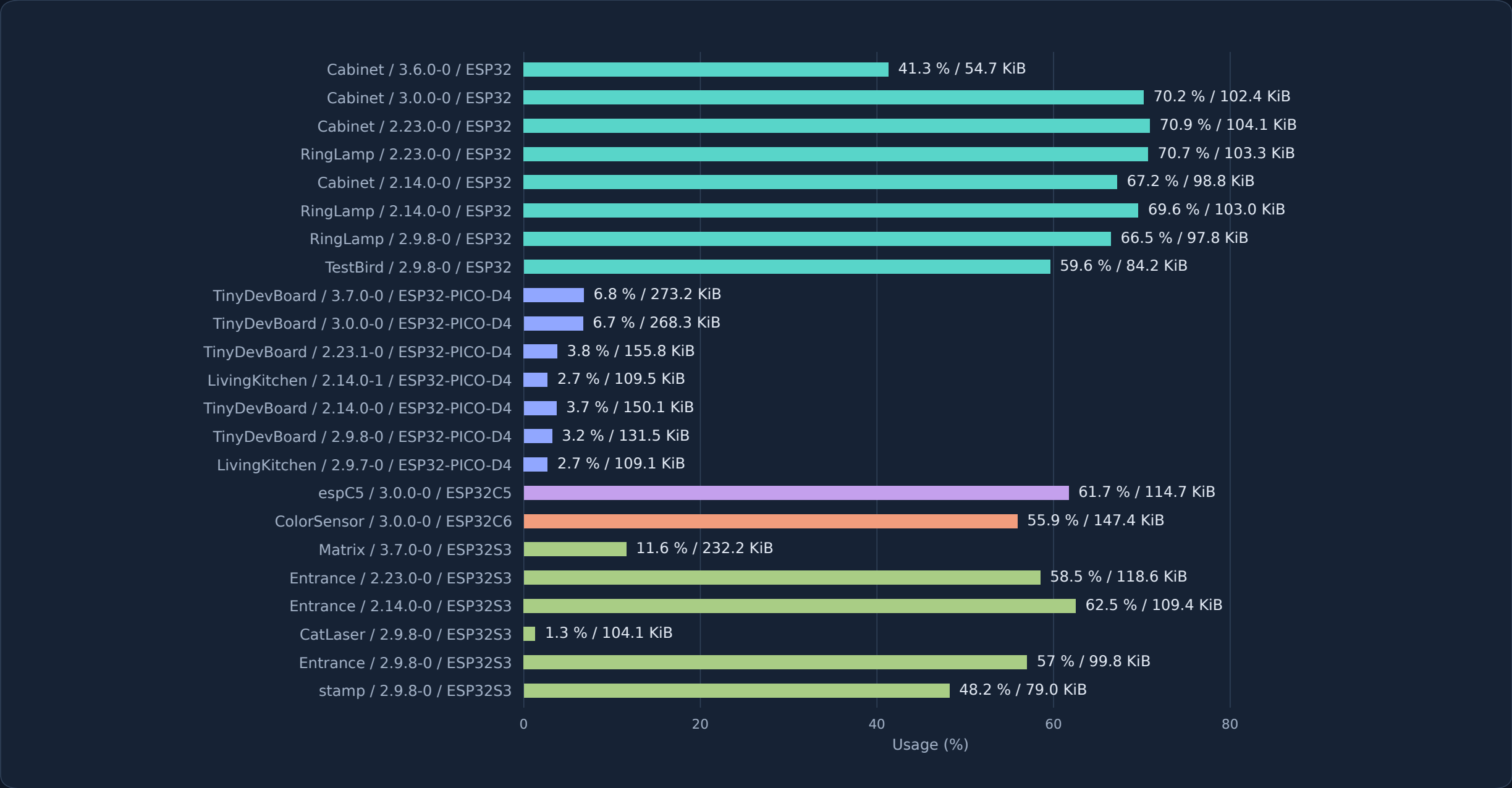
Web rest response

Device benchmarks / 23 records / newest versions first within each device type



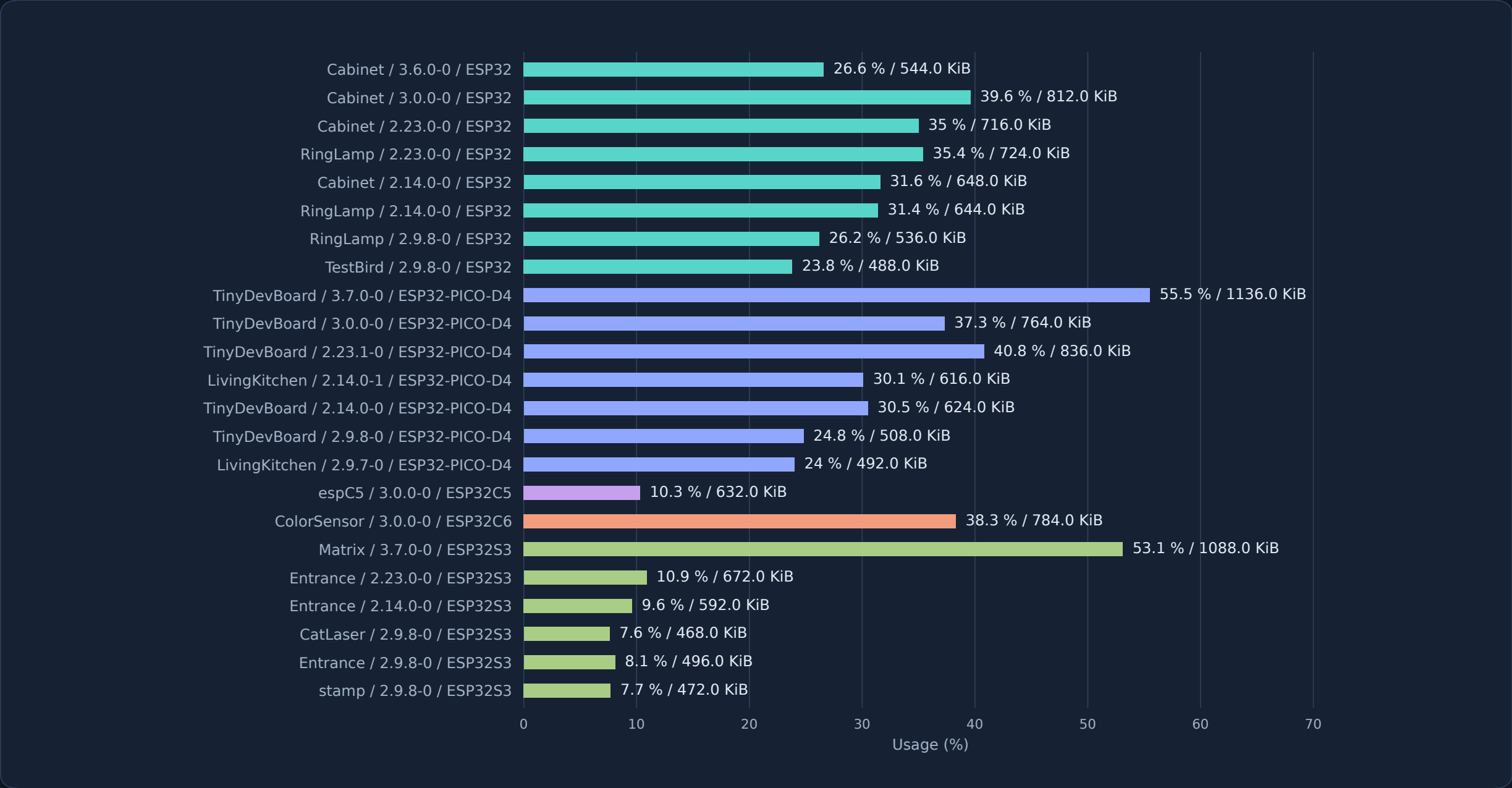
Memory utilization

Device benchmarks / 23 records / newest versions first within each device type



Filesystem utilization

Device benchmarks / 23 records / newest versions first within each device type



Device module inventory

Complete loaded-module lists for each benchmark record

Device / version	Device type	Loaded modules
Cabinet / 3.6.0-0	ESP32	web, cct, system, dht22, task
Cabinet / 3.0.0-0	ESP32	system, cct, dht22, task
Cabinet / 2.23.0-0	ESP32	system, cct, dht22, task
RingLamp / 2.23.0-0	ESP32	neopixel, system, task
Cabinet / 2.14.0-0	ESP32	system, cct, dht22, task
RingLamp / 2.14.0-0	ESP32	neopixel, pacman, system, task
RingLamp / 2.9.8-0	ESP32	neopixel, pacman, system, task
TestBird / 2.9.8-0	ESP32	types_demo, system, task
TinyDevBoard / 3.7.0-0	ESP32-PICO-D4	system, web, rgb, fileserver, cct, cluster, telegram, dimmer, oled, oled_ui, gameOfLife, task
TinyDevBoard / 3.0.0-0	ESP32-PICO-D4	gameOfLife, oled_ui, web, fileserver, dimmer, telegram, system, rgb, cluster, cct, oled, task
TinyDevBoard / 2.23.1-0	ESP32-PICO-D4	system, rgb, telegram, cct, pacman, cluster, dashboard_be, oled, oled_ui, gameOfLife, task
LivingKitchen / 2.14.0-1	ESP32-PICO-D4	rgbcct, dimmer, rgb, cct, dashboard_be, system, task
TinyDevBoard / 2.14.0-0	ESP32-PICO-D4	system, rgb, pacman, cct, telegram, dashboard_be, oled, oled_ui, gameOfLife, task
TinyDevBoard / 2.9.8-0	ESP32-PICO-D4	system, rgb, telegram, cct, dashboard_be, intercon, oled_ui, oled, gameOfLife, task
LivingKitchen / 2.9.7-0	ESP32-PICO-D4	esp32, rgbcct, dimmer, rgb, cct, dashboard_be, system, intercon, task

Device module inventory

Complete loaded-module lists for each benchmark record / continued

Device / version	Device type	Loaded modules
espC5 / 3.0.0-0	ESP32C5	system, task
ColorSensor / 3.0.0-0	ESP32C6	web, tcs3472, system, cluster, neopixel, fileserver, task
Matrix / 3.7.0-0	ESP32S3	esp32, web, system, cluster, neomatrix, fileserver, task
Entrance / 2.23.0-0	ESP32S3	buzzer, telegram, system, pacman, task
Entrance / 2.14.0-0	ESP32S3	pacman, buzzer, system, telegram, task
CatLaser / 2.9.8-0	ESP32S3	roboarm, telegram, system, dashboard_be, servo, switch, task
Entrance / 2.9.8-0	ESP32S3	telegram, system, buzzer, pacman, task
stamp / 2.9.8-0	ESP32S3	system, task

Version commit history

Latest 60 entries / newest first / full commit messages

Version	Commit ID	Message
3.7.0-0	5f78d86d28ad281774160f8ca5fb81ac2fb1f851	[micrOS][Core] Guard imported LM commands - version: 3.7.0-0
3.6.4-0	645d6f17d7c14782523e83d6d3e620b41f1d37d4	[micrOS][Core][Web] Persist web app settings - version: 3.6.4-0
3.6.3-1	c0b27f7988570fa27ce3a8514bfe49172df236d0	[micrOS][Web Config] Fix config UI drafts and Scheduler layout - 3.6.3-1
3.6.3-0	a903b75c81faffe79f9a96c8d83426ae5ef3b58b	[micrOS][WebConfig] Lazy-load Config Packages UI and mark installed packages
3.6.2-0	0699b389b7a5709a098cb3237b19daa48006cce5	micrOS/packages update
3.6.1-0	211417389d067b9826f8d950ed50f50d4ff418d8	[micrOS][webUI] Config/Scheduler updated configuration cards - version: 3.6.1-0
3.6.0-0	2fa75484b640a7d716240aaffebc27abb102f09f	[micrOS][Core][PYPI] Add memory-aware WebUI defaults - version: 3.6.0-0
3.5.0-0	ca8f0dc0062bb4bbe301f283df3d5c83d369d69e	[micrOS][Web UI - Files] Improve fileserver UI and editor workflow
3.4.2-0	e6d22b9dc02053e1251a7ba4799207facaddbf99	[micrOS][Core] Harden web routing and frontend authentication - 3.4.2-0
3.4.1-0	2908010f3a267279d09fb75a3cf0f229b5da3cac	[micrOS][Web] Landing page web UI redesign - version: 3.4.1-0
3.4.0-1	c82376cae0663042a6107c386ddc0781c2e942da	Move LM_OV2640.py, LM_Id2410.py, LM_roboarm.py into packages + fix devToolkit --lint to include packages in core ref counts too
3.4.0-0	ccab5bed1904a5f631ba5e689e71b04b010d7af2	Config grouping visual update
3.3.2-0	3090979bdc9e1735c4831860c496d85a0e9e9de1	[micrOS][Core][Web] Return stream outputs directly and add Debug diagnostics UI - version: 3.3.2-0
3.3.1-0	d464b8e2c5544689848cdd15b849eea746dc3469	Minimal web config UI update (Pin-mapping -> Pinmap), and package update (irrigation_system)
3.3.0-0	f0a1e38df86b53b560841deed51f6262d33d2006	[micrOS][LoadModule] Add protected config reboot endpoint

Version commit history

Latest 60 entries / newest first / full commit messages / continued

Version	Commit ID	Message
3.2.5-0	6cb0490381d7b22d79537b178acebb9751ed379b	[micrOS][PYPI] Web Config editor updates - version: 3.2.5-0
3.2.4-0	cc496a2a5b29ae02381d0ce46c374ad95dd056ee	[micrOS][WebConfig] Polish web UI controls and mobile REST input - 3.2.4-0
3.2.3-0	10336d15c21c71dd42ab4d14bf9ad9c443da4c18	[micrOS][WebConfig] Normalize scheduler and REST config UI - version: 3.2.3-0
3.2.2-0	5a0646f33a7cbc01a11a9d031ba93ebaa5c77979	[micrOS] Improve config web UI controls - version: 3.2.2-0
3.2.1-0	4c7464edeb2aadee61716934c8b5e8a8174d94bf	[micrOS][LoadModule] Improve fileserver directory metadata and mobile UI - version: 3.2.1-0
3.2.0-0	5c5c8c486e7b84b371cd8c96b3e1cf8479fb7bab	Config WebUI - sfuncman_3.2.0-0.json - version: 3.2.0-0
3.1.1-0	ca733296be5cc20fdff08c835d3cfea588d581ca	[micrOS] move LM_co2.mpy and LM_mh_z19c.py into co2 package, remove them from default LMs + package submodule update
3.1.0-0	79209de34dc7f59f7e607ab50b62597de6282d91	[devToolKit][ToolKit] Optimize web resources during cross compile + event based dashboard widgets - version: 3.1.0-0
3.0.10-0	49ff95b16ec2c367bbc864f883c9d64ef7a45282	Refactor modules into (installable) packages: LM_VL53L0X.py, LM_sound_event.py, LM_tcs3472.py
3.0.9-0	1204dfe4f925a3635401a0ef6afd9c73ce4142cb	[micrOS] Minor fixes in Files.py and LM_cluster.py, LM_pacman.py improvement (pack_ls command)
3.0.8-0	17916d676e005bb50ff9dac7ee873979af8a1f9d	[micrOS] Add uapi_widgets.js (modules helper auto load) for autocomplete
3.0.7-0	f7d45a039056ac2aafe279c5b5086e22e899398d	[micrOS][PYPI] web: split optional UI plugins and improve editor UX - version: 3.0.7-0
3.0.6-0	87d69a40f364f46040296dae1ba4be372e1efaaf	[micrOS] web config ui beta endpoints (/config (GET/POST), /config/ui (GET)) in LM_web.py + config.html template
3.0.5-0	d08dcc36f605d070001330e356f317b4220b196d	[micrOS] Web dashboard optimization and LM_OV2640 cam module integration

Version commit history

Latest 60 entries / newest first / full commit messages / continued

Version	Commit ID	Message
3.0.4-0	f68555d913254d05b5b3130f17f156c2d0584a07	[microOS][PYPI] Config.py (read/write) improvements + utests/test_config.py + faster test_scheduler.py - version: 3.0.4-0
3.0.0-0	6faf9bb256bf882156f084f65e8caac11a8ef095	[microOS] Micro-improvements - string literal reduction
2.30.0-0	9a82cf056dc2695a5202de031ab8c117860db741	[microOS] HA and analysis updates - version: 2.30.0-0
2.29.0-0	7d0b7f3d882d160d73e83fb1eab0ed52efff673c	[microOS] Performance and stability improvements
2.28.0-0	82de41362c421f13b65e4d98bb8feffdd2d68ca5	[microOS] Boot sequence flow memory optimization and WDT integration into microOS task.feed (30 sec) - initialized from microOS.py - version: 2.28.0-0
2.27.2-0	3e07063afb6cdf5321093ae274a35f2eeb8aa81	Doc update microOS/ARCHITECTURE.md + APPLICATION_GUIDE.md -> microOS/MODULE_GUIDE.md
2.27.1-0	9a1fe1fa1a9c46b0c6033ecc7ca2bd5c87ce8a5f	[microOS] Fix unsupported media types issue in fs/editor for py, dat, cache, log types -> fallback to plain/text as txt - version: 2.27.1-0
2.27.0-0	9f9810051bd1e05d0e907786af6bc1bc44777e4c	[microOS] Fix simulator usecases (fs-editor is broken...) - version: 2.27.0-0
2.26.3-0	b68437c641d48f509f73cdb66814a3598e17a857	[microOS] Web UI enhancements (ustyle.css) - version: 2.26.3-0
2.26.1-0	bdab6d5927c16f2735e910aca22ca9b66307fb76	[microOS][PYPI] Minor improvements - LM_neomatrix.py migration under microSPackages - minor tldr improvements - release candidate (3.0) - version: 2.26.1-0
2.26.0-0	8ae5ee3ecc24a4ea7b061f2cf786ffe4dd41b1dd	[microOS] Small fix for LM_OV2640.py.photo() endpoint
2.25.3-0	247a27860eb8002edc0eb4076707c83e7cfafaae	[microOS][PYPI] 3.0.0 Candidate - Pacman.py (LM_pacman.py) package upgrade feature version: 2.25.3-0
2.25.2-1	c8e0df15a4c9a43ce8de2968ee72607cbf2993b6	LM_OV2640.py photo endpoint - filesaver improvement: can take multiple photos with timestamp up to storage 60% - filserver usage callback reuse in OV2640 (PoC)

Version commit history

Latest 60 entries / newest first / full commit messages / continued

Version	Commit ID	Message
2.25.2-0	30d2651194defeb1364a2bf1bc85aaaa49e68be2	[micrOS] Web Homepage (index.html) visual improvements: endpoint groups :)
2.25.1-0	bdad9ac20f27b3f44d290983ec1f8c383376e053	[micrOS] Web.py generic plain/text fallback (only check /web root file extensions) + OV2640 fileserver integration with /cam/photo endpoint
2.25.0-0	f24b37fa53b1835d1693f19195967a61bcb06cec	[micrOS][PYPI] Web.py->web_endpoints enhancement - version: 2.25.0-0
2.24.1-0	0c0b66499f0376795be4680c842d56bcabfa1350	[micrOS] simulator_lib/uasyncio.py wait_for() fix + micrOS Files WebUI beta and fileserver enhancements - version: 2.24.1-0
2.24.0-0	611f1757de64201305699c13b904acc75bff098e	Doc update and updated precomiled files
2.23.1-0	e7e6ba28fd51be3d5ee96b17a25e5ad5a3fe87d5	[micrOS] Espnow.py less strict remote command execution (restrct unloaded modeules execution with auth:True in config) + Fix dashboard UI node status visualization - version: 2.23.1-0
2.23.0-0	4c29abe36f2933f476f6845d252bbaf9852ba9b7	[micrOS][PYPI] Fix USB configuration issues (backup/restore) + remove legacy telegram features, from now telegram is only a LoadModule with micrOS system Notify integration - version: 2.23.0-0
2.22.1-0	718fc75f721ecd7588ed449f81b5368481a12f8	[micrOS] InterConnect, Espnow improvement - version: 2.21.3-0
2.22.0-0	014d55d7a2ee120a2031656d0694cf20e8bd8c8a	[devToolKit] toolkit/socketClient.py embedding fix into devToolKit.py -c with -p param fixes
2.21.4-0	103166148a01a6057b8ffd65ef74ccac7d8edb07	[micrOS] Pacman.py.install local mip registry unpack support quick fix and micrOS/packages update
2.21.3-0	6844ebcdf42a1902a67a93d7bd43711195f86fdd	[DevToolKit] New micrOS/packages --unpack handles 3PP packages, toolkit/simulator_lib/simulator.py: Monkey-patch time module duw to 3PP dependencies, remove hardcoded toolkit/simulator_lib/mqtt_as dependency
2.21.2-0	0db0f84878ce0096bf9056cd5b5a21de4e4145f8	[micrOS] microlO.py and LM_system.py pinmap update, show avaiialble pinmap names as well

Version commit history

Latest 60 entries / newest first / full commit messages / continued

Version	Commit ID	Message
2.21.1-0	cb3ce11a779c718f1466b8ca73d511412040df4d	[microOS] WebEngine refactor with new dimensioning solution (is_limited and chunking size adjustments): >20kb No webUI, 20-100kb use default chunking settings, <100kb upscale chunking size to enhance file tansaction speed - version: 2.21.1-0
2.21.0-0	31e5b9a34100415cbf5ba2053d0ae273990b0e70	[microOS] REPL helper/tools: helpers.py: reset and shell shortcuts, shell makes possible to run ShellCli interface over UART (REPL) (same as socket ShellCli) BETA
2.20.2-0	d33cc344de4ec0703db7a733ef86411d180fdbd1	ESPNow: module name added to espnow task ID
2.20.1-0	484d1d813649413512b75c1a2d641377e622a307	ESPNow: transaction ID implementation and refactor
2.20.0-0	dd5060453b49c41f0579fc43c6337526dbc45e65	[microOS] Unified task (micro_task) creation response: {'task_tag': 'verdict'} - version: 2.20.0-0
2.19.0-0	ed7aef655490199721f9f65be19746c8ea312ad4	[microOS][PYPI] InterConnect.py logging fix, socketClient.node_status feature status addition - version: 2.19.0-0